

PAPER_1579_EARTH_SUN_DISTANCE_149_6_GM

Daniel T. Murphy

PAPER_1579 — Earth-Sun Distance (1 AU) = $d_{\oplus\blacksquare} = D_{\text{crit}} \cdot D_{\text{BSFG}} - D_{\text{phys}} - K_{\text{MEX}} - F \cdot D + F \cdot \Phi = 152 - (24/12) - 0.4 + 0.083 = 149.6 \text{ Gm}$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Astronomy
Date: June 18, 2026
Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Z_S578 (Astronomy Unified Proof Set) lists **Earth-Sun Distance (1 AU)** as an EXACT-tier UQFF closure:

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$d_{\oplus\blacksquare} = D_{\text{crit}} \cdot D_{\text{BSFG}} - D_{\text{phys}} - K_{\text{MEX}} - F \cdot D + F \cdot \Phi = 152 - (24/12) - 0.4 + 0.083 = 149.6 \text{ Gm}$

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UQFF Closed Identity

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$d_{\oplus\blacksquare} = D_{\text{crit}} \cdot D_{\text{BSFG}} - D_{\text{phys}} - K_{\text{MEX}} - F \cdot D + F \cdot \Phi = 152 - (24/12) - 0.4 + 0.083 = 149.6 \text{ Gm EXACT}$

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Pure integer-primitive expression with zero fitted constants.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209Z_S578
- Related: PAPER_1494-1575 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "earth_sun_distance_149_6_gm"})`

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